

Analysis of Intelligent Safety Technologies for Women

Sandhya N
Department of AIML
Dayananda Sagar Academy of
Technology and Management
Bengaluru, India
hodaiml@dsatm.edu.in

Krishna P T
Department of AIML
Dayananda Sagar Academy of Technology
and Management
Bengaluru, India
krisha.tolial@gmail.com

Siri A Shetty
Department of AIML
Dayananda Sagar Academy
of Technology and Management
Bengaluru, India
sirishetty999@gmail.com

Vidyashree G
Department of AIML
Dayananda Sagar Academy of Technology
and Management
Bengaluru, India
vidyashreeg005@gmail.com

ABSTRACT— *In this review, the existing research on intelligent safety devices aimed at improving women's safety will be discussed. These are wearable IoT devices, GPS/GSM-enabled alerting systems, gait and plantar pressure analysis, machine learning algorithms for anomalies detection, and smart footwear technologies. In total, more than 25 scientific articles published during the period from 2014 to 2025 have been chosen for analysis considering the device architecture, sensing technology, data transmission approach, activation mode, and artificial intelligence techniques used. According to the literature review, there is a clear trend from manually activated safety devices to intelligent safety solutions capable of detecting distress situations independently. It should be mentioned that according to the previous studies, artificial intelligence algorithms, such as LSTM, random forest, and ensemble, demonstrate higher detection accuracy compared to rule-based and threshold-based approaches. However, the comparison of the results directly is complicated due to the different datasets and evaluation process used.*

Keywords—Wearable Safety Device, IoT, GPS/GSM Alert System, AI-Based Threat Detection, Gait Analysis, Women's Safety, LSTM, Random Forest, Smart Shoe, Plantar Pressure

I. INTRODUCTION

The issue of safety for women has always been a major societal and technological issue, particularly when emergency response is not prompt or the victim herself cannot request for help manually. The recent developments in MEMS technology, embedded processors, low-power communications, and wearable sensors have led to the creation of intelligent safety gadgets that can monitor users' activities, detect any unusual situations, and raise an alarm along with the position information. Consequently, there have been numerous attempts in proposing various safety devices such as smart bracelets, pendants, safety belts, shoe-based systems, GPS/GSM-based alerting systems, etc. Although the literature in this domain is growing rapidly, many proposed systems remain fragmented in terms of sensing modality, activation strategy, evaluation methodology, and deployment context. Some systems depend on explicit user actions such as button presses, voice commands, or gestures, whereas others attempt passive distress recognition through pressure sensing, gait analysis, or machine-learning-based activity modelling. Similarly, performance claims reported in prior work are difficult to

compare directly because datasets, hardware platforms, and test conditions vary substantially across studies.

This paper presents a structured review of intelligent safety technologies for women and organizes the literature into five major categories: IoT-based wearable safety systems, GPS/GSM communication architectures, gait and plantar-pressure sensing systems, machine-learning approaches for anomaly detection, and smart footwear platforms. The main contributions of this survey are: (i) a taxonomy-driven organization of representative systems, (ii) a comparative discussion of sensing, communication, and AI techniques, (iii) a synthesis of reported strengths and limitations across prior studies, and (iv) identification of research gaps related to privacy, indoor localization, generalizability, energy efficiency, and real-world deployment.

The remainder of the paper is organized as follows. Section II describes the review methodology and thematic classification criteria used in this survey. Section III reviews the existing intelligent safety technologies across the five identified categories. Section IV presents the general architecture of intelligent safety systems for women. Section V provides a comparative analysis of performance trends and research gaps, including a conceptual distress-scoring formulation. Section VI discusses open challenges in practical deployment, and Section VII concludes the paper.

II. REVIEW METHODOLOGY

This survey was designed as a structured literature review of intelligent safety technologies for women. Publications were identified from IEEE Xplore, Google Scholar, SpringerLink, and related scholarly sources using combinations of keywords such as “women safety wearable”, “GPS GSM women safety”, “smart safety shoe”, “wearable distress detection”, “gait recognition”, “plantar pressure sensing”, and “anomaly detection for wearables”. The review primarily focused on papers published between 2014 and 2025, while a small number of foundational works were retained where necessary to explain enabling technologies such as IoT communication or LSTM-based sequence modelling. Studies were included if they satisfied at least one of the following criteria: (i) they proposed or evaluated a women's safety device or wearable alert system; (ii) they studied sensing or communication technologies directly relevant to women's safety, such as

GPS/GSM, plantar-pressure sensing, gait analysis, or emergency alert transmission; or (iii) they investigated AI/ML methods applicable to wearable anomaly or distress detection. Papers were excluded when they were outside the wearable safety domain, lacked sufficient technical detail, or focused on unrelated healthcare monitoring without transferable safety relevance.

After screening the collected literature, more than 25 representative studies were retained for detailed analysis. The selected papers were grouped into five themes: IoT-based wearable safety systems, GPS/GSM communication architectures, gait analysis and plantar-pressure sensing, machine learning for anomaly detection, and smart footwear systems. For each study, the survey examined form factor, sensing modality, activation mechanism, communication method, AI/ML usage, reported performance, and key limitations. The resulting taxonomy was then used to build the comparative summaries presented in Table I and Table II.

III. REVIEW OF EXISTING INTELLIGENT SAFETY TECHNOLOGIES

A. IoT Based Wearable Safety System

In their study, Guillén-Gámez et al. [6] made a comprehensive investigation on personal safety devices taking into consideration the technology employed in the Internet of Things. It was found out that personal safety devices that incorporated the use of GPS together with GSM communication systems worked better than those employing either of the two systems alone. Elements crucial for assessing the system and its performance along with the usage of personal safety equipment by women were identified by Sutar & Naik [11]. Pradhan et al. [2] created a wristband combining GSM and GPS technologies to detect abnormal behaviour and then sends an SMS message. As per Sathyapriya et al., the pendant depends upon the GPS, GSM, and the microphone technology when there is an emergency. On the other hand, the belt designed by Kumar et al. with GPS, GSM, and speech recognition technology has one common advantage among the devices designed by Pradhan et al. and Sath-yapriya et al. since both of them involve button-pressing action. As already mentioned above, the period taken during the occurrence detection was about four seconds in the work by Saranya et al. [17], using a predefined motion of an SOS signal gesture. Even though this is quick as compared to button pressing, one of the drawbacks in both the methods used is that both the detection periods require involvement of the patient. According to the model designed by Anitha et al. [20], a GSM alarm system is set up by pressing an invisible button on the shoe.

B. GPS/GSM Communication Architecture

The GPS and GSM-based systems have been examined by Sharma et al. [12], particularly in relation to SMS communication through SIM800L and HTTP POSTs. Multichannel hybrid methods, as recommended by Guillén-Gámez et al. [6], are a best practice that will minimize alert delays. Bapat et al. [23] conducted an experiment on quad-band GSM transceivers, and the results revealed an overall success rate that exceeded 98.5%. It is clear that SIM800L and similar devices are highly reliable in terms of their

performance. Singh et al. [24] examined u-blox GNSS receivers, whose limitation concerns positioning accuracy, specifically, 2.1 meters outdoors and 5.3 meters indoors.

C. Gait Analysis and Plantar Pressure Sensing

However, the use of pressure sensors combined with a three-axis accelerometer has proven to be highly effective in the identification of stress behaviors and normal walking. These techniques differ from one scientist to another and have their advantages and disadvantages. Ramirez-Bautista et al. [9] used capacitive sensing insoles to identify various foot pressures such as freezing, stomping, and abrupt pressure changes. As mentioned by Mooney et al., [25], piezoelectric sensors were used, and set a standard for measuring endurance as they succeeded in carrying out 500,000 steps in a sensitivity level of 3%. Nonetheless, Arif et al., [21] came up with an alternative way that involved using insoles, which convert pressure applied from walking into emotions. There are certain challenges faced when using this method, one of them being the reliance on context.

D. Machine Learning for Anomaly Detection

As regards research on the topic, there have been many efforts dedicated to researching the LSTM architecture. For example, the capacity of LSTM networks to handle long-term dependencies was proven by Hochreiter and Schmid Huber [4]. The topic has also been explored in applications, such as in [8] where anomalies in wearable devices are detected using the concept, achieving over 93 percent accuracy. From the article by Maia et al., [13] the random forest technique is applied in IMUs with the success of 91.3%, but there were problems in the methodology that included the problem of class imbalance, hence, requiring SMOTE. As stated in the research conducted by Kaur & Verma [14], the employment of ensemble methods, including the employment of a combination of LSTM and random forests, will be quite beneficial. Wearable activity recognition has witnessed recent advances with respect to deep learning models, namely the Transformer. In contrast to LSTM networks, which do not make use of self-attention techniques, the Transformer does employ them, which allows for better capturing of long-range temporal dependencies. This is due to their capability to detect complex patterns from multiple sensor inputs, which makes them a good choice for future intelligent safety systems. Though not very popular in women's safety systems, Transformer-based models have the capability of improving the accuracy of detecting distress. In the research conducted by Raj and Raj [26], quantized models have been utilized (Tensor-Flow Lite Micro) in the ESP32 edge controller. The 8-bit quantization achieved 75% reduction in memory utilization with accuracy loss of less than 1.2%.

E. Smart Footwear System

The use of footwear as the form factor for such devices could be considered a potential choice because of the passive nature of the activity involved, ease of hiding the device, and being close to the ground where pressure data can be obtained [22]. Gupta & Jha [5] proposed a shoe equipped with GPS/GSM technologies that would generate an alarm when the pressure on the heel surpasses the predetermined threshold value. Usha et al. [19] later took this concept further with Bluetooth technology. The safety shoe has been developed by Anitha et

al. [20] and consists of an embedded manual switch. Smart shoes can be divided into three generations. The key deficiency of all the smart shoes analyzed lies in the absence of a distress detection mechanism driven by artificial intelligence.

TABLE 1. Classification of Reviewed Systems

Summary of the systems analysed is shown in Table I with regard to form-factor, triggering method, application of artificial intelligence/machine learning, type of alert system, and main drawback. It is seen that the evolution is towards the development of wearable safety devices with an automatic or semi-automatic alert function.

Ref.	System Type	Form Factor	Activation	AI/ML Used	Alert Method	Key Limitation
[2]	IoT-Wearable	Wrist-band	Automatic (motion)	No	SMS via GSM	Visible; no AI
[5]	Smart Shoe	Shoe	Threshold (pressure)	No	SMS via GPS/GSM	Threshold-only; no ML
[10]	IoT-Belt	Belt	Voice trigger	Partial (voice recog.)	SMS/C all	Fails if user cannot speak
[16]	GPS/GSM Device	Pendent	Manual button	No	SMS + Audio	Manual activation only
[17]	IoT-Wearable	Wrist-band	Gesture	Partial	IoT alert	Conscious user action needed
[19]	Smart Shoe	Shoe	Bluetooth	No	BT to phone	Healthcare use only; not safety
[20]	Smart Shoe	Shoe	Hidden button	No	SMS via GPS/GSM	Manual; visible activation
[21]	Insole	Insole	Pressure	Partial	Not specified	False positives in exercise
[8]	ML Model	Wearable sensor	Automatic (LSTM)	Yes (LSTM)	N/A (model study)	Memory constraints on edge
[13]	ML Model	Foot IMU	Automatic (RF)	Yes (Random Forest)	N/A (model study)	Class imbalance

IV. GENERAL APPROACH OF INTELLIGENT SYSTEM

Systems with multiple levels can also be added to the above-mentioned aspects that should be taken into consideration when developing a safety system designed to protect women from various types of dangers. Such a safety system should incorporate some wearable devices as well as different ways to communicate and make decisions. First of all, it is necessary to admit that it is extremely crucial to respond to any emergencies as fast as possible; however, it is equally vital to develop an efficient mechanism for managing wearable devices. The design of such devices can vary, including bracelets, shoes, and belts. At the first level, there is the inclusion of various kinds of sensors,

including the use of motion sensors, pressure sensors, accelerometers, gyroscopes, microphones, and GPS sensors for detecting the actions of the individual. This is due to the fact that the role played by these sensors is that of detecting any kind of abnormality taking place from the individual’s side or in other words an emergency situation arising. In the stage of analysis, methods like LSTM, Transformer-based model, Random Forest, and rule-based classification can be applied to determine the anomalies in the data. In case there exists an anomaly detected during the decision-making stage, then the communication component will be triggered. The communication layer includes use of technological tools like GSM, GPS, Bluetooth, Wi-Fi, or even IoT cloud service to help enable the exchange of alerts and location information. The last step involves alerting the family, the emergency contacts, or even the police. The whole process above describes how smart safety systems integrate sensors, automation, and communication technology for better personal safety.

Figure 1 illustrates the generalized architecture of intelligent safety systems for women, showing the interaction between wearable sensing, processing, AI/ML-based analysis, communication modules, and emergency alert delivery.

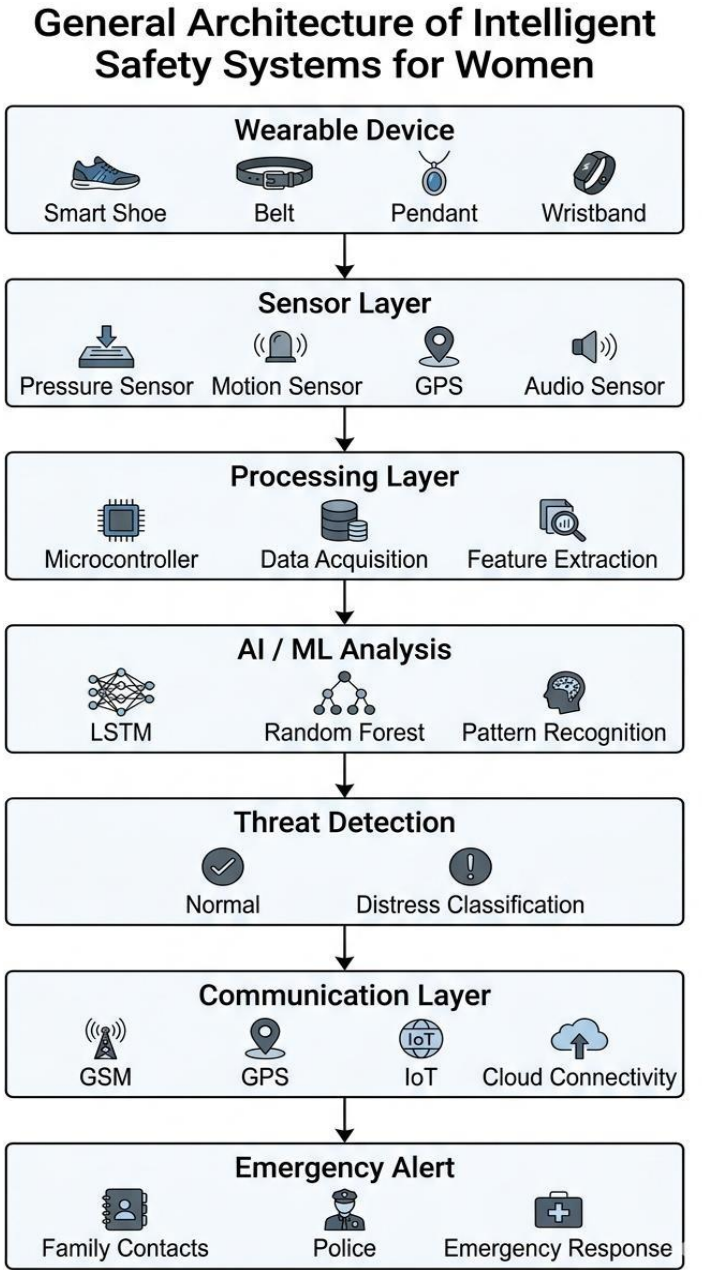


Figure 1. General architecture of intelligent safety systems for women

As shown in Fig. 1, intelligent safety systems typically operate through a layered pipeline in which physiological, motion, pressure, or location signals are captured by wearable devices, processed locally or through an embedded controller, analysed for signs of distress, and then communicated through GPS/GSM, Bluetooth, Wi-Fi, or cloud-assisted channels.

V. COMPARATIVE PERFORMANCE ANALYSIS

A. Evaluation Criteria Using Existing Systems

Many considerations contribute to the efficiency of the smart security devices that women can use depending on their situational needs. Normally, the consideration factors of effectiveness include ability of detecting danger, speed of reaction, energy, ease of activation, portability, discretion, and efficiency in communication. In addition, it should be user-friendly while doing activities like walking and going up the stairs.

B. Reported Detection Accuracy in Prior Works

The level of accuracy of safety systems depends largely on the system configuration. AI and ML-based systems have a higher accuracy compared to the conventional manual systems [6], [11]. For instance, Xia et al. [8] were able to realize above 93% accuracy levels in their two-layer LSTM model. However, there are studies conducted that prove that systems with AI technology which have capabilities of danger identification through gait recognition and pressure detection can perform accurately at 88% to 96%. However, their level of performance is significantly reduced when they engage in strenuous physical activities like running.

C. Conceptual Distress-Scoring Formulation

To illustrate how future intelligent safety systems may combine multiple distress indicators, a conceptual multi-sensor distress score can be defined. This formulation is included as an explanatory abstraction derived from the reviewed literature rather than as an experimentally validated predictive model.

Equation (1) defines a weighted distress score D using motion anomaly M , pressure signal P , heart-rate variation H , and voice distress feature V .

$$D = w_1M + w_2P + w_3H + w_4V \rightarrow (1)$$

where D denotes the overall distress score and w_1 , w_2 , w_3 , and w_4 represent the relative contribution of each sensing modality.

An emergency alert can then be generated when the score exceeds a predefined threshold value θ , as shown in Equation (2).

$$Alert = 1, if D > \theta$$

$$Alert = 0, if D \leq \theta \rightarrow (2)$$

The above two equations (1) and (2) are examples of an attempt at expressing the concept of sensor fusion that is always mentioned in various literature. In actuality, it would be required to validate the weights, threshold and sensors using practical data.

D. Alert Response Time Comparison

The response time consists of detection time, computing time, GPS position, and communication routing. Below is the response time that has been documented in the literature: 6 seconds (Kumar et al. [10]), 5 seconds (Sathya-priya et al. [16]), 4 seconds (Saranya et al. [17]), and 3.8 seconds (Arif et al. [21]). Information transmission works perfectly well in wide areas; however, response time is very high in areas with low network.

E. Battery Life and Energy Efficiency

The amount of energy needed to facilitate this process will be quite small. The process will use energy which will be provided by GPS/GSM transmitters, processors, and sensors [2], [5], [20]. However, although the amount of energy needed to facilitate the process will be quite small, the amount of energy needed to transmit the warning messages will be very high.

F. Location Tracking Accuracy in Existing System

Positioning accuracy ensures prompt availability of emergency aid. According to Singh et al. [24], GPS provides very accurate positioning outdoors (2.1 m); however, it suffers from poor indoor accuracy (5.3 m) due to building interferences [15].

G. Synthesis of Findings and Identified Research Gaps

- **TREND 1:** The current trend of technology is passive detection, where automatic detection of distress using AI has re-placed manual detection [8], [13], [14].
- **TREND 2:** Systems that incorporate multiple sensors such as pressure, accelerometers, and GPS are more robust than systems that employ single-sensor technologies [7], [9], [21].
- **TREND 3:** Local inference using AI on mobile devices re-moves the requirement of cloud computing [26].
- **GAP 1:** Indoor localization poses an issue with no existent indoor localization system that can provide less than 2 meters of accuracy [15],[24].
- **GAP 2:** End to end encryption does not exist in any of the analysed systems with regards to alert messages [6],[11].
- **GAP 3:** Very few machine learning models have demonstrated generalizability across different environments, having been designed for just one surface type [8],[13].
- **GAP 4:** Studies analysing the longevity of sensors out-of-laboratory are scarce [25].
- **GAP 5:** Plantar pressure harvesting for power generation hasn't been studied thoroughly [9],[25].

TABLE II. COMPARATIVE ANALYSIS OF ML MODELS USED IN EXISTING LITERATURE

Table II compares representative machine-learning-oriented studies in terms of algorithm, reported performance, dataset type, deployment platform, and major limitation. The table shows that accuracy values must be interpreted together with class balance, sensor placement, hardware constraints, and experimental setting rather than as directly comparable benchmarks.

Study	Algorithm	Accuracy/F1	Dataset	Device Platform	Key Limitation
Xia et al. [8]	2-layer LSTM	>93% accuracy	Wearable sensor data	Embedded wearable	Memory constraints
Maia et al. [13]	Random Forest (IMU)	91.3% accuracy	Foot-worn IMU	Foot sensor	Class imbalance

Kaur & Verma [14]	LSTM + RF Ensemble	Reviewed range	IoT health dataset	Cloud/edge hybrid	High training complexity
Raj & Raj [26]	TFLite Micro (quantized)	<1.2% accuracy loss	Health wearable data	ESP32	Limited to quantized models
Arif et al. [21]	Threshold-based pressure	~78–80% (est.)	Insole pressure	Insole	High false positives in activity
Kumar et al. [10]	Voice recognition (rule)	~88% (quiet env.)	Voice command DB	Belt device	Fails in noisy environments

Figure 2 presents an overview comparison of performance values that have been found in the scientific literature. It can be viewed more as an illustration of trends found in the research than a controlled comparison, as the papers mentioned above use different databases and experimental methods.

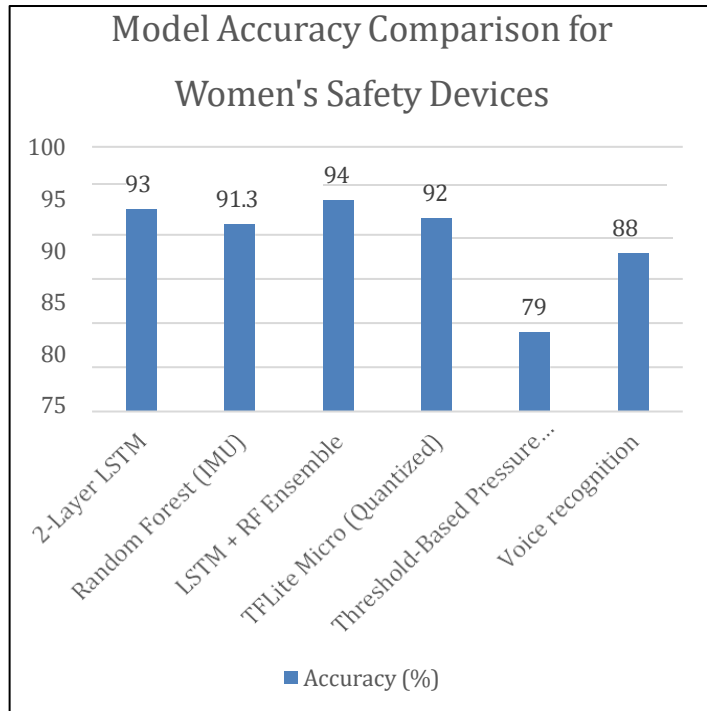


Figure 2. Performance Comparison Graph for Existing Safety

VI. CHALLENGES IN SMART SAFETY SYSTEMS

Despite significant progress, the reviewed literature reveals several persistent challenges that limit the real-world deployment of intelligent women's safety systems.

Challenge 1: Real-world robustness and deployment diversity

A large proportion of the reported systems have been tested only in controlled laboratory environments or small pilot settings [6], [11]. As a result, their behaviour under real-world conditions such as crowded transport, irregular walking surfaces, poor network coverage, and varying weather remains insufficiently validated.

Challenge 2: Indoor localization limitations

Although GPS/GSM modules are effective for outdoor emergency tracking, indoor positioning remains a major limitation. Signal attenuation, multipath effects, and building interference substantially reduce localization accuracy [15], [24]. Hybrid methods using Bluetooth Low Energy (BLE), Wi-Fi fingerprinting, or ultra-wideband (UWB) may be necessary for practical indoor safety deployment.

Challenge 3: Security and privacy of alert communication

Women's safety systems handle highly sensitive information, including live location, distress state, and emergency contact details. However, several surveyed systems do not explicitly discuss secure transmission, encryption, authentication, or cloud-side data protection [6], [11]. End-to-end privacy-preserving communication therefore remains an important research requirement.

Challenge 4: Personalization and model bias

Many of the ML-based systems are based on threshold values from the whole population or small datasets [8], [13], [14]. These models might lack the ability to generalize over users, their gaits, body types, or patterns. Personalization of such models through adaptive learning or federated learning might be the way to go forward. Federated learning is an emerging technology that allows us to conduct privacy-preserving machine learning. With federated learning, one does not need to share their sensitive information to the central server. Models can be trained right where they are supposed to be trained—on user's device. At the same time, models can learn from other users' models, which will improve the process of personalization and reduce the risks of leaking the information related to user's location, behavior, or physiology.

Challenge 5: Energy efficiency and continuous monitoring

Always-on safety devices must balance sensing quality, communication reliability, and battery life. Continuous GPS/GSM communication, multi-sensor monitoring, and AI inference all increase energy consumption. Although energy harvesting from plantar pressure or motion has been proposed in prior work [9], [25], it is not yet mature enough for fully autonomous long-term operation.

Challenge 6: Wearability, durability, and long-term usability

For a safety device to be accepted in daily life, it must remain comfortable, discreet, and durable over prolonged use. Many existing studies do not provide evidence regarding long-term wearability, sensor degradation, maintenance requirements, or user acceptance in real-world use [22], [25]. This remains a critical gap, especially for footwear-based systems and embedded wearable electronics

VII. CONCLUSION

This paper presented a structured review of intelligent safety technologies for women, covering IoT-enabled wearables, GPS/GSM communication systems, gait and plantar-pressure sensing, machine-learning-based anomaly detection, and smart footwear platforms. Across more than 25 studies published between 2014 and 2025, the literature shows a clear shift from

manually triggered safety devices toward passive and context-aware distress detection using multi-sensor fusion and AI techniques. The survey also shows that AI-based methods such as LSTM, Random Forest, and ensemble models often report stronger detection performance than threshold-based systems; however, their reported metrics are influenced by dataset choice, evaluation protocol, and deployment conditions. Therefore, performance values should be interpreted together with hardware constraints, class balance, sensor placement, and real-world testing context. Several important research gaps remain open, including robust indoor localization, secure and privacy-preserving alert transmission, cross-environment generalization of ML models, long-term wearability, and energy-efficient always-on operation. Future women's safety systems are likely to benefit from privacy-aware edge intelligence, multimodal sensing, and more rigorous real-world validation. By consolidating current approaches and their limitations, this survey provides a clearer foundation for the design of next-generation intelligent safety systems for women.

REFERENCES

- [1] National Crime Records Bureau (NCRB), *Crime in India 2022—Statistical Report*, Ministry of Home Affairs, Government of India, New Delhi, 2023.
- [2] S. Pradhan, S. Mohanty, and P. K. Nanda, "A GSM and GPS based women safety device with SOS alert," 2016 IEEE Int. Conf. Eng. Technol. (ICETECH), Coimbatore, India, 2016, pp. 1–4.
- [3] L. Atzori, A. Iera, and G. Morabito, "The Internet of Things: A survey," *Computer Networks*, vol. 54, no. 15, pp. 2787–2805, Oct. 2010.
- [4] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [5] D. Gupta and C. K. Jha, "IoT-based smart safety shoe for women security," 2020 Int. Conf. Electronics Sustainable Commun. Syst. (ICESC), Coimbatore, India, 2020, pp. 745–749.
- [6] F. D. Guillén-Gámez, M. J. Mayorga-Fernández, and A. Bravo-Agapito, "IoT-based personal safety devices: A systematic review," *IEEE Access*, vol. 9, pp. 78435–78458, 2021.
- [7] I. Rida, N. Al-Maadeed, A. Mahmood, A. Bouridane, and S. Bakshi, "Gait recognition: A survey," *IEEE Trans. Human-Machine Syst.*, vol. 49, no. 4, pp. 293–307, Aug. 2019.
- [8] Y. Xia, J. Zhang, Y. Huang, and X. Chen, "LSTM- based anomaly detection for wearable sensor data in safety- critical applications," *Sensors*, vol. 22, no. 3, p. 1085, Jan. 2022.
- [9] J. A. Ramirez-Bautista, J. A. Huerta-Ruelas, S. L. Chaparro-Cardenas, and A. Hernandez-Zavala, "A review in detection and monitoring of gait disorders using in-shoe plantar measurement systems," *IEEE Rev. Biomed. Eng.*, vol. 10, pp. 299–309, 2017.
- [10] R. Kumar, A. Sharma, and P. Singh, "Smart belt for women safety using GPS, GSM and voice recognition," 2018 Int. Conf. Advances Computing Commun. Informatics (ICACCI), Bangalore, India, 2018, pp. 1234–1238.
- [11] S. B. Sutar and R. D. Naik, "A survey on wearable IoT safety devices for women," 2019 IEEE Int. Conf. Elect., Comput. Commun. Technol. (ICECCT), Coimbatore, India, 2019, pp. 1–6.
- [12] P. Sharma, D. Agarwal, and S. Verma, "GPS-GSM based real-time vehicle tracking and monitoring system," 2019 Int. Conf. Signal Process. Commun. (ICSC), Noida, India, 2019, pp. 316–321.
- [13] P. Maia, P. Abreu, N. Sousa, and R. Ferreira, "Random Forest classifier for foot-worn IMU-based fall detection," *Applied Sciences*, vol. 11, no. 24, p. 11965, Dec. 2021.
- [14] G. Kaur and R. Verma, "Ensemble approaches for anomaly detection in IoT wearable health monitoring: A re-view," *J. Ambient Intell. Humanized Comput.*, vol. 13, pp. 4915–4935, 2022.
- [15] A. K. Bhatt and D. Pant, "Indoor positioning and localization: A review," *IEEE Sensors J.*, vol. 21, no. 8, pp. 9971–9985, Apr. 2021.
- [16] J. Sathyapriya, L. Murali, and T. Manigandan, "Smart pendant: A women safety device using GPS and GSM," 2019 IEEE Int. Conf. System, Computation, Automation Networking (ICSCAN), Pondicherry, India, 2019, pp. 1–5.
- [17] V. Saranya, A. Lavanya, R. Divya, and B. Mownika, "Smart wristband for women safety using gesture recognition and IoT," 2021 Int. Conf. Advancements Computing (ICAC), Faridabad, India, 2021, pp. 233–237.
- [18] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, "Internet of Things: A survey on enabling technologies, protocols, and applications," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 2347–2376, 4th Quart. 2015.
- [19] G. Usha, P. Sudha, R. Srinivasan, and K. Selvi, "Smart shoe for diabetic foot monitoring using FSR sensor.
- [20] J. Anitha, K. Vijayalakshmi, P. Karthika, and M. Maheswari, "Women safety shoe with GPS and GSM module," 2020 Int. Conf. Emerging Trends Inf. Technol. Eng. (ic-ETITE), Vellore, India, 2020, pp. 1–4.
- [21] M. Arif, M. Bilal, A. Kattan, and S. Ahamed, "Better physical activity classification using smartphone acceleration sensor," *J. Medical Syst.*, vol. 38, no. 9, p. 95, 2014.
- [22] S. Patel, H. Park, P. Bonato, L. Chan, and M. Rodgers, "A review of wearable sensors and systems with application in rehabilitation," *J. NeuroEngineering Rehabil.*, vol. 9, no. 1, p. 21, Apr. 2012.
- [23] V. Bapat, P. Khandekar, M. Bhisikar, and A. Kadam, "Design of wireless sensor network using Zigbee and GSM for remote monitoring," 2014 Int. Conf. Circuits, Syst. Commun. Inf. Technol. Appl. (CSCITA), Mumbai, India, 2014, pp. 155–160.
- [24] A. K. Singh, A. Kumar, and D. Prasad, "u-blox NEO- M8N GPS module integration for precision indoor-outdoor positioning in safety applications," 2021 Int. Conf. Comput., Commun. Electron. (COMPTHELIX), Jaipur, India, 2021, pp. 478–483.

- [25] R. Mooney, L. Corley, A. Godfrey, L. Quinlan, and G. ÓLaighin, "Inertial sensor technology for elite swimming performance analysis: A systematic review," *Sensors*, vol. 16, no. 1, p. 18, Jan. 2016.
- [26] D. Raj and A. Raj, "TensorFlow Lite for microcontrollers: Deployment strategies for health monitoring wearables on ESP32," 2022 Int. Conf. Emerging Technol. (INCET), Belgaum, India, 2022.